**PARIS-AFRICANA INTERNATIONAL SCHOOL**
**NO. 5 PARIS AFRICANA STREET, OFF TANKO AL-MAKURA ROAD**
**VIA SANI ABACHA ROAD, MARRABA**
**CLASS: SS3**

## THEORY PART

**ANSWER ALL QUESTIONS**

1. It is known that 2 out of every 5 cigarettes smokers is a village have cancer of the lungs. Find the probability that out of a random sample of 8 smokers from the village, 5 will have cancer of the lungs.

b. 20% of the total production of transistors produced by a machine are below standard. If a random sample of 6 transistors produced by the machine is taken, what is the probability of getting
   (i) exactly 2
   (ii) at least 2
   (iii) at most 2 standard transistors?

2. A sphere has a volume of $1000cm^3$
   i. Calculate its radius correct to 3.s.f.2.
   ii. Find the surface area of the sphere.

b. A hemisphere bowl has an external radius of 24cm and is made of wood 3cm thick calculate the volume of wood in the bowl.

3. Given that Q = $\begin{pmatrix} 6 & 0 \\ 4 & 5 \end{pmatrix}$ $and\ Q + P = \begin{pmatrix} 7 & -2 \\ 6 & 8 \end{pmatrix}$ , evaluate $|Q + 2P|$.

4. Express $\frac{3\sqrt{5}-\sqrt{3}}{2\sqrt{5}+3\sqrt{3}}$ in the form a√15 +b, where a and b are rational numbers.

b. Evaluate the following

(i) $\frac{3}{\sqrt{3}}\left(\frac{2}{\sqrt{3}} - \frac{\sqrt{12}}{6}\right)$ (ii) $\frac{\sqrt{75}}{\sqrt{48}-\sqrt{3}}$

5. Simplify $\frac{3}{4\sqrt{6}+5}$

b. Simplify $\frac{5}{\sqrt{7}+\sqrt{3}} + \frac{7}{\sqrt{7}-\sqrt{3}}$

6. Resolve into partial fractions
   a. $\frac{3x-1}{(x+1)(x-2)}$
   b. $\frac{x+12}{2x^2+3x-2}$

7. Given that $\log 3 = 0.4771,\ \log 2 = 0.3010,$ find the values of the following without using logarithm table.
   i. $\log\sqrt{6}$
   ii. $\log\sqrt[3]{0.3}$

8. If y= $4x^2$-3x differentiate with respect to x

9. Calculate the Simple interest on #55000 for 5 years at 7% per annum.

10. Calculate the Compound interest on #200000 for 3 years at a rate of 2.5%.

ENGLISH LANGUAGE –

Section A – Composition (Choose ONE)

Write NOT LESS than 450 words on any one of the following:

1. A Memorable Day I Will Never Forget
2. The Effects of Social Media on Teenagers Today
3. A Speech to Your School Prefects on How to Improve Discipline
4. "Examination Malpractice Is the Bane of Our Educational System." – Write an article.
5. Write a story that ends with:

"…and that was when I realised the importance of telling the truth."

Section B – Summary

Read the passage below and answer the questions.

PASSAGE

Many students perform poorly in examinations not because they are unintelligent, but because they lack proper study habits. Effective study requires planning, discipline, and consistency. Some students wait until the examination period before they begin to prepare, and this often leads to tension, confusion, and poor performance. Others rely on copying in the examination hall, but this habit only weakens their confidence and prevents real learning.

Good study habits involve setting aside regular time for reading, taking notes, asking questions in class, and revising lessons weekly. Students who practise these habits often understand their subjects better and feel more confident during examinations. Teachers, on their part, should guide students on how to study properly. Parents should also create a quiet environment at home and encourage their children by showing interest in their academic progress.

When students, teachers, and parents work together, better academic performance becomes possible. Success in examinations is not a matter of luck but the result of hard work, discipline, and proper guidance.

QUESTIONS

1. In two sentences, state the reasons why many students perform poorly in examinations.
2. Mention two good study habits highlighted in the passage.
3. In one sentence, state the role teachers are expected to play.
4. In one sentence, state how parents can assist students.
5. What is necessary for better academic performance?

Section C – Collocations

Write the correct collocations from the brackets to complete the sentences.
1. The students were advised to ___ (take notes / make noise) during the lesson.
2. She hopes to ___ (make progress / do harm) in her studies this term.
3. Parents should ___ (make effort / give up) to support their children academically.
4. The principal will ___ (hold a meeting / break a record) with all prefects tomorrow.
5. He failed the test because he did not ___ (pay attention / take revenge) in class.

Section D – Verb Forms

Fill in the correct verb form in each sentence.
1. By this time next year, we ___ (finish) our WAEC examinations.
2. She usually ___ (go) to school early, but today she was late.
3. They ___ (write) the test when the teacher walked in.
4. The students have ___ (submit) their assignments already.
5. If he had studied harder, he ___ (pass) the exam.